Technical Note TN-07 – Cavitation in centrifugal pumps

Cavitation occurs when the pressure at the impeller eye falls below the vapour pressure of the liquid. Vapour bubbles form and then collapse violently in higher-pressure zones, causing noise, vibration, loss of head and pitting of the impeller.

1. Symptoms

- Crackling noise, like gravel passing through the pump.
- Fluctuating discharge pressure and motor current.
- Increased vibration (on MD-Pompes panels: code P-50 may follow code P-41).
- Pitting on impeller vanes at inspection.

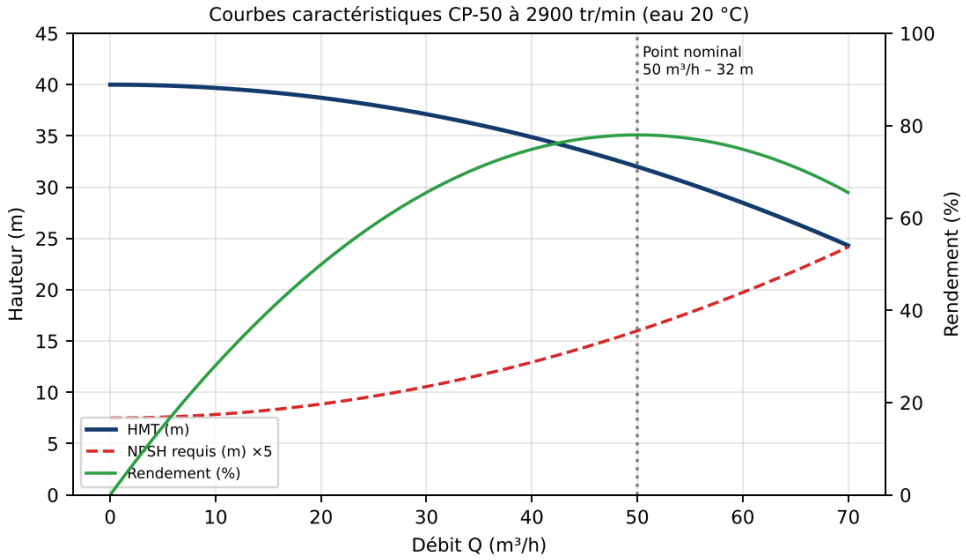


Figure 1 – NPSH required rises with flow rate (CP-50 example)

2. Root causes and remedies

Root cause	Remedy
Clogged suction strainer	Clean strainer (see POMPE-B-01 history, 2025-08)
Partially closed suction valve	Fully open suction valve; throttle only on discharge
Low tank level	Restore level; check LSL-101
Operation far right of the curve	Reduce flow to ≤ 110 % of nominal
Hot liquid	Check NPSH available with actual temperature

Table 1 – Causes and remedies

NOTE – Rule of thumb: keep NPSH available at least 0.5 m above NPSH required.